K- Nearest Neighbor Algorithm

Find a phoneme for test frame using the Euclidean distance measure.

/a/ :	2	4	5	Training Data
/a/ :	3	5	6	
/i/ :	4	5	3	
/i/ :	3	4	5	
/i/ :	2	3	4	

Test Data : 2 5 4

Answer:

From Training Data :

$dist_a^2 = (2-2)^2 + (4-5)^2 + (5-4)^2 = 2$ *

$dist_a^2 = (3-2)^2 + (5-5)^2 + (6-4)^2 = 5$

$dist_i^2 = (4-2)^2 + (5-5)^2 + (3-4)^2 = 5$

$dist_i^2 = (3-2)^2 + (4-5)^2 + (5-4)^2 = 3$ *

$dist_i^2 = (2-2)^2 + (3-5)^2 + (4-4)^2 = 4$ *

if $K = 3$,

Minimum are $dist_a^2$, $dist_i^2$, $dist_i^2$

If we take majority voting,

One vote for a

Two vote for i

Winner is i

Output phoneme = /i/

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/a/ :	2	4	5	Training data
/a/ :	3	5	6	
/i/ :	4	5	3	
/i/ :	3	4	5	
/i/ :	2	3	4	
Test data :	2	5	4	

Answer:

Training Data:

For /a/ : Mean calculation

$$\mu_{1a} = \frac{2+3}{2} = 2.5, \quad \mu_{2a} = \frac{4+5}{2} = 4.5, \quad \mu_{3a} = \frac{5+6}{2} = 5.5$$

For /i/ : Mean calculation

$$\mu_{1i} = \frac{4+3+2}{3} = 3, \quad \mu_{2i} = \frac{5+4+3}{3} = 4, \quad \mu_{3i} = \frac{3+5+4}{3} = 4$$

Testing Data:

Using Euclidean distance measure

$$d_a^2 = (\mu_{1a} - 2)^2 + (\mu_{2a} - 5)^2 + (\mu_{3a} - 4)^2$$
$$= (2.5-2)^2 + (4.5-5)^2 + (5.5-4)^2 = 2.75$$

$$d_i^2 = (\mu_{1i} - 2)^2 + (\mu_{2i} - 5)^2 + (\mu_{3i} - 4)^2$$
$$= (3-2)^2 + (4-5)^2 + (4-4)^2 = 2$$

Since $d_i^2 < d_a^2$, Output phoneme = /i/

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